77. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I	List-II
A. Potassium bromide	1. Fertilizer
B. Potassium nitrate	2. Photography
C. Potassium sulphate	3. Bakery
D. Monopotassium tartarate	4. Gun-power

Codes:

	A	B	C	D
(a)	2	4	1	3
(b)	2	4	3	1
(c)	4	2	3	1
(d)	4	2	1	3

Ans. (a) : Potassium bromide is used in photography as a restrainer in black and white developer formulas. Gun powder, also called black powder, is a mixture of sulphur, charcoal, and potassium nitrate. Gun powder can be made by just using potassium nitrate and charcoal (or alternatively without charcoal), but without the sulphur (or coal), the powder is not as strong. Potassium sulfate is primarily used as a fertilizer. Mono potassium tartrate is used in bakery by combination with baking soda it results in to evolution of CO2 which is used for baking cakes.

78. Which one of the following elements is essential for the construction of nuclear reactors?

(a) Cobalt (b) Nickel
(c) Zirconium (d) Tungsten

Ans. (c) : Zirconium is a chemical element with symbol Zr and atomic number 40. The name zirconium is taken from the name of the mineral zircon, the most important source of Zirconium. Zirconium is used in nuclear reactors to provide the dadding or outer covering for the cylindrical fuel rods that power a nuclear reaction.

79. Consider the following statements:

Coke is one of the materials of the charge added to blast furnace for the production of steel/iron. Its function is to:

1. Act as the reducing agent
2. Remove silica associated with the iron ore
3. Function as fuel, to supply heat
4. Act as an oxidizing agent

Of these statements–

(a) 1 and 2 are correct
(b) 2 and 4 are correct
(c) 1 and 3 are correct
(d) 3 and 4 are correct

Ans. (c) : Coke is used as a charge in the iron-producing furnace, which aids in the blasting in the iron furnace. Coke is used as a reducing agent and fuel in an iron furnace. Coke is not used in iron ore to separate mixed silica (sand) and to act as an oxidant.

80. Which one of the following metals does not form amalgams?

(a) Zinc (b) Copper
(c) Magnesium (d) Iron

Ans. (d) : Amalgam is an alloy consisting of mercury and any other element. Iron being exceptional in nature not form amalgam with mercury. That's why mercury is stored in vessels made up of iron. Small quantities of an iron amalgam have, however, been formed by immersing sodium amalgam (containing 1 percent sodium) in a clear, saturated solution of ferrous sulphate, While rest three options zinc, copper and magnesium combine with mercury to form there respective amalgam.

81. Consider the following statements about acetylene:

1. It is used in welding industry.
2. It is a raw material for preparing plastics.
3. It is easily obtained by mixing silicon carbide and water.

Of these statements

(a) 1 and 2 are correct
(b) 1 and 3 are correct
(c) 2 and 3 are correct
(d) 1, 2 and 3 are correct

Ans. (a) : Welding refers to the process of joining two or more metals together. Approximately 20% of acetylene is consumed for oxyacetylene gas welding and cutting due to high temperature of flame. Acetylene can be semihydrogenerated to form ethylene, providing a feedstock for a variety of Polyethylene plastics. When silicon carbide and water is mixed it produces Silica and methane. Hence 3rd statement is incorrect.

BIOLOGY

82. Which one of the following types of microorganisms is most widely used in Industries?

(a) Bacteria
(b) Bacteria and Fungi
(c) Bacteria and Algae
(d) Bacteria, Microalgae and Fungi

Ans. (d) : Bacteria, micro-algae and fungi are most widely used in industries.

83. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I	List-II
A. Fruit	1. Ovule
B. Seed	2. Leaf
C. Wood	3. Stem
D. Starch	4. Ovary

Codes:

	A	B	C	D
(a)	2	1	3	4
(b)	4	1	3	2
(c)	2	3	1	4
(d)	4	3	1	2

Ans. (b) : The correct match is as follows:

List-I		List-II	
A.	Fruit – Ovary	1.	(Ripen ovary of a plant)
B.	Seed – Ovule	2.	(Fertilized ovule develop into a seed)
C.	Wood – Stem	3.	(Wood can be obtained from stem of a tree)
D.	Starch – Leaf	4.	(Formed as a result of photosynthesis by leaves)

84. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I	List-II
A. Theory of Mutation	1. Beadk and Tatum
B. Theory of Evolution	2. Jacob and Monod
C. One gene one enzyme hypothesis	3. Darwin
D. Operon concept	4. Devries

Codes:

	A	B	C	D
(a)	3	4	1	2
(b)	4	3	1	2
(c)	4	3	2	1
(d)	3	4	2	1

Ans. (b) : The theory of mutation was formulated by Hugo de vries. Charles Darwin is credited with proposing the theory of the evolution of organisms. The concept of one gene one enzyme was proposed by George Beadle and Edward Tatum. The operon concept was proposed by Francois Jacob and Jacques Monod.

85. Which one of the following statements regarding starch and cellulose is NOT correct?

(a) Both of them are of plant origin
(b) Both of them are polymers
(c) Both of them give colour with iodine
(d) Both of them are made up of glucose molecules

Ans. (c) : Starch is the reserve substance in plant cells whereas cellulose is the most important structural component of the cell wall of plants. Both starch and cellulose are polymers of glucose. In contrast to starch and glycogen, cellulose is insoluble in ordinary solvents and is not hydrolysed by boiling dilute acids. It gives no colour with iodine.

86. Ergotism is due to consumption of:

(a) Contaminated grains
(b) Rotting vegetables
(c) Contaminated water
(d) Stale cooked food

Ans. (a) : Ergotism is the effect of long term ergot poisoning, due to ingestion of alkaloids produced by fungus Claviceps purpurea which is found in infected cereals and ryes.

87. The complete conversion of glucose in the presence of oxygen into carbon dioxide and water with release of energy is called:

(a) Aerobic respiration
(b) Anaerobic respiration
(c) Glycolysis
(d) Hydrolysis

Ans. (a) : Aerobic respiration takes place in the presence of oxygen and finally degrades glucose to carbon dioxide and water with release of energy in form of ATP.

88. The major chemical compound found in human kidney stones is:

(a) Uric acid
(b) Calcium carbonate
(c) Calcium oxalate
(d) Calcium sulphate

Ans. (c) : A kidney stone or renal calculus is also known as nephrolith is a solid material formed in the kidneys from the minerals found in the urine. While most of the cases of kidney stones, this is caused due to genetic reasons, environmental factors or a combination of the two, factors such as obesity, intake of certain foods, medications and lack of water intake also attribute to the formation of kidney stones. The major chemical compound found in human kidney stones is calcium oxalate or oxalate of lime. With the chemical formula CaC_2O_4 or $Ca(COO)_2$.

89. Consumption of fish is considered to be healthy when compared to flesh of other animals because fish contains:

(a) Polyunsaturated fatty acids
(b) Saturated fatty acids
(c) Essential vitamins
(d) More carbohydrates and proteins

Ans. (a) : Polyunsaturated fatty acid found in fish is an anti oxidant, which helps in building of tissues.

90. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I (Disease)	List-II (Organism)
A. Malaria	1. Fungi
B. Poliomyelitis	2. Bacteria
C. Tuberculosis	3. Virus
D. Tingworm	4. Protozoane